



Curso de Tecnologia em Sistemas para Internet
Disciplina de Fundamentos de Sistemas Computacionais
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TRABALHO FINAL

Exercício A: abra o firefox, um editor de texto e vários outros programas e depois execute o comando top. veja como as trocas entre processos são muito dinâmicas. agora, responda em detalhes: o que o cabeçalho do comando mostra e o que cada coluna do comando significa? para cancelar o comando utilize a combinação de teclas ctrl + c. lembrando que:

ctrl + c → encerra o processo em execução.

ctrl + z → suspende um processo.

```
top - 19:21:37 up 16 min, 2 users, load average: 1,40, 1,02, 0,81
Tarefas: 366 total, 3 em exec., 363 dormindo, 0 parado, 0 zumbi
%CPU(s): 0,8 us, 0,6 sy, 0,0 ni, 98,6 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 6296,4 free, 4447,5 used, 6146,8 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used, 11232,8 avail mem

  PID  USUARIO  PR  NI  VIRT  RES  SHR  S  %CPU  %MEM  TEMPO+  COMANDO
 2268  utfpr    20  0  5381132 345768 154604 S   7,8   2,2   0:50.06  gnome-shell
 4305  utfpr    20  0  1395,7g 596392 144428 S   1,8   3,7   1:21.71  chrome
 7699  utfpr    20  0  706464 68080 54008 R   1,8   0,4   0:00.44  gnome-terminal-
 6744  utfpr    20  0  755984 188204 119856 S   1,4   1,2   0:02.91  software-proper
 6553  utfpr    20  0  1585116 149680 105052 R   0,9   0,9   0:01.21  gnome-system-mo
 131   root     0 -20    0      0      0 D   0,5   0,0   0:01.12  kworker/u49:0+i915_flip
 167   root     0 -20    0      0      0 I   0,5   0,0   0:00.22  kworker/11:1H-i915_cleanup
 644   root     20  0      0      0      0 I   0,5   0,0   0:00.65  kworker/u48:13-i915
 2476  utfpr    20  0  386716 12988 7604 S   0,5   0,1   0:01.66  ibus-daemon
 7745  utfpr    20  0  12124 6300 3968 R   0,5   0,0   0:00.03  top
 1     root     20  0  24024 15456 9712 S   0,0   0,1   0:01.96  systemd
 2     root     20  0      0      0      0 S   0,0   0,0   0:00.00  kthreadd
 3     root     20  0      0      0      0 S   0,0   0,0   0:00.00  pool_workqueue_release
 4     root     0 -20    0      0      0 I   0,0   0,0   0:00.00  kworker/R-rcu_gp
 5     root     0 -20    0      0      0 I   0,0   0,0   0:00.00  kworker/R-sync_wq
 6     root     0 -20    0      0      0 I   0,0   0,0   0:00.00  kworker/R-kvfree_rcu_reclaim
 7     root     0 -20    0      0      0 I   0,0   0,0   0:00.00  kworker/R-slub_flushwq
 8     root     0 -20    0      0      0 I   0,0   0,0   0:00.00  kworker/R-netns
top - 19:21:50 up 16 min, 2 users, load average: 1,18, 0,99, 0,81
Tarefas: 366 total, 1 em exec., 365 dormindo, 0 parado, 0 zumbi
```

Resposta: As primeiras linhas mostram como o seu computador está lidando com a carga geral, Tempo e Carga, tarefas (Tasks), porcentagem da CPU e Memória. E a lista dinâmica foca em quem está consumindo o quê.

Exercício B: Crie dois novos usuários pedro e madalena. Depois, alterne para o usuário pedro e digite o comando top no terminal dele. Novamente, alterne para o usuário madalena e digite top no terminal dele. Por fim, alterne para o seu usuário e digite top no seu terminal. Perceba as atividades em tempo real dos diferentes usuários do SO Linux e seus processos. Faça um print de tela salvando-o. Você pode fazer isso alternando o usuário utilizando a parte gráfica do Sistema Operacional.

```
utfpr@GP-LAB4-W1003:/home$ sudo adduser pedro
[sudo] senha para utfpr:
info: Adicionando usuário 'pedro' ...
info: Selecionando UID/GID da faixa 1000 a 59999 ...
info: Adicionando novo grupo 'pedro' (1002) ...
info: Adicionando novo usuário 'pedro' (1002) com grupo 'pedro (1002)' ...
info: Criando diretório pessoal '/home/pedro' ...
info: Copiando arquivos de '/etc/skel' ...
Nova senha:
SENHA INCORRETA: A senha não passa pela verificação do dicionário - é muito simples/sistemática
Redigite a nova senha:
passwd: senha atualizada com sucesso
Modificando as informações de usuário para pedro
Informe o novo valor ou pressione ENTER para aceitar o padrão
  Nome Completo []:
  Número da Sala []:
  Fone de Trabalho []:
  Fone Residencial []:
  Outro []:
A informação está correta? [S/n]
info: Adicionando novo usuário 'pedro' a grupos suplementares / extras 'users' ...
info: Adicionando usuário 'pedro' ao grupo 'users' ...
```

```
utfpr@GP-LAB4-W1003:/home$ pwd
/home
utfpr@GP-LAB4-W1003:/home$ su pedro
Senha:
pedro@GP-LAB4-W1003:/home$ whoami
pedro
pedro@GP-LAB4-W1003:/home$
```

```

top - 19:37:41 up 32 min, 2 users, load average: 0,21, 0,26, 0,44
Tarefas: 352 total, 2 em exec., 350 dormindo, 0 parado, 0 zumbi
%CPU(s): 3,8 us, 1,0 sy, 0,0 ni, 95,2 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 7526,0 free, 3267,2 used, 5832,2 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used. 12413,1 avail mem

```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TEMPO+	COMANDO
4305	utfpr	20	0	1392,5g	414112	149036	R	12,3	2,6	1:59.70	chrome
2268	utfpr	20	0	5346928	351808	144324	S	9,0	2,2	1:27.97	gnome-shell
14380	utfpr	20	0	3964048	149096	102460	S	7,6	0,9	0:00.23	gnome-character
14376	utfpr	20	0	939672	56924	44652	S	4,7	0,4	0:00.14	nautilus
14382	utfpr	20	0	499992	34956	27516	S	2,3	0,2	0:00.07	gnome-control-c
6596	utfpr	20	0	1854096	155492	100668	S	2,0	1,0	0:00.66	gnome-clocks
1905	utfpr	20	0	11520	7324	4804	S	1,0	0,0	0:00.56	dbus-daemon
2476	utfpr	20	0	386716	13004	7604	S	1,0	0,1	0:03.39	ibus-daemon
128	root	20	0	0	0	0	I	0,3	0,0	0:00.31	kworker/6:1-i915-unordered
383	root	19	-1	67260	21136	19444	S	0,3	0,1	0:00.45	systemd-journal
677	root	20	0	0	0	0	I	0,3	0,0	0:01.12	kworker/u48:29-i915
1370	syslog	20	0	222572	6588	4632	S	0,3	0,0	0:00.11	rsyslogd
1876	utfpr	20	0	461864	11356	10096	S	0,3	0,1	0:00.11	gnome-keyring-d
2206	utfpr	20	0	312040	8668	7756	S	0,3	0,1	0:00.05	gvfsd
2705	utfpr	20	0	419804	31304	19764	S	0,3	0,2	0:02.03	ibus-extension-
2756	utfpr	20	0	234632	8128	7320	S	0,3	0,1	0:01.10	ibus-engine-sim
2848	utfpr	39	19	806720	31972	19548	S	0,3	0,2	0:00.37	tracker-miner-f
3053	utfpr	20	0	533552	78988	58124	S	0,3	0,5	0:00.48	Xwayland
4176	root	0	-20	0	0	0	I	0,3	0,0	0:02.13	kworker/u49:3-i915_flip
7699	utfpr	20	0	708484	70396	54136	S	0,3	0,4	0:03.73	gnome-terminal-
12961	root	20	0	0	0	0	I	0,3	0,0	0:00.26	kworker/u48:0-events_unbound
14370	pedro	20	0	12144	6236	3944	R	0,3	0,0	0:00.03	top
1	root	20	0	24024	15456	9712	S	0,0	0,1	0:02.04	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/0:0H-events_highpri
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	kworker/0:1-i915-unordered
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0,0	0,0	0:00.08	ksoftirqd/0
15	root	20	0	0	0	0	I	0,0	0,0	0:02.68	rcu_preempt
16	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.01	migration/0

```

pedro@GP-LAB4-W1003:/home$ pwd
/home
pedro@GP-LAB4-W1003:/home$ su madalena
Senha:
madalena@GP-LAB4-W1003:/home$ whoami
madalena
madalena@GP-LAB4-W1003:/home$ top

```

```

top - 19:39:21 up 34 min, 2 users, load average: 0,45, 0,33, 0,44
Tarefas: 353 total, 1 em exec., 352 dormindo, 0 parado, 0 zumbi
%CPU(s): 4,2 us, 1,1 sy, 0,0 ni, 94,4 id, 0,2 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 7477,3 free, 3306,5 used, 5838,4 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used, 12373,8 avail mem

```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TEMPO+	COMANDO
2268	utfpr	20	0	5338784	353376	144324	S	11,6	2,2	1:34.93	gnome-shell
14748	utfpr	20	0	3964048	149836	103384	S	8,3	0,9	0:00.25	gnome-character
14744	utfpr	20	0	1005184	56332	44568	S	6,0	0,4	0:00.18	nautilus
6596	utfpr	20	0	1861120	162744	100668	S	4,3	1,0	0:00.88	gnome-clocks
1905	utfpr	20	0	11532	7336	4804	S	1,0	0,0	0:00.65	dbus-daemon
2877	utfpr	20	0	3218252	74756	55820	S	0,7	0,5	0:01.06	gjs
4305	utfpr	20	0	1392,5g	424996	149220	S	0,7	2,6	2:05.23	chrome
1	root	20	0	24024	15456	9712	S	0,3	0,1	0:02.06	systemd
172	root	20	0	0	0	0	I	0,3	0,0	0:02.00	kworker/11:1-events
383	root	19	-1	67260	20952	19260	S	0,3	0,1	0:00.47	systemd-journal
1859	utfpr	20	0	21732	13152	9768	S	0,3	0,1	0:00.61	systemd
1876	utfpr	20	0	461864	11368	10096	S	0,3	0,1	0:00.13	gnome-keyring-d
2476	utfpr	20	0	386716	13004	7604	S	0,3	0,1	0:03.68	ibus-daemon
2478	utfpr	20	0	411340	21892	18812	S	0,3	0,1	0:00.11	gsd-color
2711	utfpr	20	0	308148	7812	7120	S	0,3	0,0	0:00.05	ibus-portal
2756	utfpr	20	0	234632	8132	7320	S	0,3	0,1	0:01.20	ibus-engine-sim
2919	utfpr	20	0	961640	15936	12932	S	0,3	0,1	0:00.17	xdg-desktop-por
2992	utfpr	20	0	48,9g	382640	253648	S	0,3	2,4	0:46.25	chrome
3055	utfpr	20	0	48,4g	183456	102520	S	0,3	1,1	0:11.63	chrome
7699	utfpr	20	0	708960	70824	54136	S	0,3	0,4	0:04.50	gnome-terminal-
14739	madalena	20	0	12144	6240	3952	R	0,3	0,0	0:00.02	top
14746	utfpr	20	0	524892	18056	15988	S	0,3	0,1	0:00.01	gnome-calculato
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/0:0H-events_highpri
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	kworker/0:1-i915-unordered
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0,0	0,0	0:00.09	ksoftirqd/0
15	root	20	0	0	0	0	I	0,0	0,0	0:02.82	rcu_preempt
16	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.01	migration/0
19	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/0

```

madalena@GP-LAB4-W1003:/home$ su utfpr
Senha:
utfpr@GP-LAB4-W1003:/home$ whoami
utfpr
utfpr@GP-LAB4-W1003:/home$ top

```

```
top - 19:41:03 up 35 min, 2 users, load average: 0,40, 0,35, 0,44
Tarefas: 357 total, 4 em exec., 353 dormindo, 0 parado, 0 zumbi
%CPU(s): 2,4 us, 0,8 sy, 0,0 ni, 96,8 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 7446,9 free, 3327,6 used, 5853,5 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used, 12352,6 avail mem
```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TEMPO+	COMANDO
2268	utfpr	20	0	5329464	353568	144324	R	9,0	2,2	1:42.51	gnome-shell
15202	utfpr	20	0	3359492	146256	102440	R	6,0	0,9	0:00.18	gnome-character
15198	utfpr	20	0	1087108	56900	44524	S	4,7	0,4	0:00.14	nautilus
15274	utfpr	20	0	434592	31520	23496	R	1,7	0,2	0:00.05	seahorse
6596	utfpr	20	0	1865892	167356	100668	S	1,3	1,0	0:00.98	gnome-clocks
4305	utfpr	20	0	1392,5g	423036	150184	S	1,0	2,6	2:13.34	chrome
2476	utfpr	20	0	386716	13004	7604	S	0,7	0,1	0:03.94	ibus-daemon
3046	utfpr	20	0	49,2g	198760	125572	S	0,7	1,2	0:50.20	chrome
7699	utfpr	20	0	709236	71080	54136	S	0,7	0,4	0:05.07	gnome-terminal-
15	root	20	0	0	0	0	I	0,3	0,0	0:02.99	rcu_preempt
129	root	20	0	0	0	0	I	0,3	0,0	0:00.22	kworker/7:1-i915-unordered
144	root	0	-20	0	0	0	I	0,3	0,0	0:00.23	kworker/0:1H-i915_cleanup
677	root	20	0	0	0	0	I	0,3	0,0	0:01.28	kworker/u48:29-kvfree_rcu_reclaim
1121	systemd+	20	0	17572	7792	6880	S	0,3	0,0	0:00.65	systemd-oomd
1870	utfpr	9	-11	113448	15028	9132	S	0,3	0,1	0:00.24	pipewire
1905	utfpr	20	0	11532	7336	4804	S	0,3	0,0	0:00.71	dbus-daemon
2280	utfpr	20	0	9620	5468	4752	S	0,3	0,0	0:00.05	dbus-daemon
2483	utfpr	20	0	410184	21472	18460	S	0,3	0,1	0:00.11	gsd-keyboard
2673	utfpr	20	0	308316	7440	6676	S	0,3	0,0	0:00.03	gvfs-gphoto2-vo
2693	utfpr	20	0	386964	8488	7380	S	0,3	0,1	0:00.12	gvfs-afc-volume
2705	utfpr	20	0	419804	31304	19764	S	0,3	0,2	0:02.15	ibus-extension-
2848	utfpr	39	19	806720	34212	19548	S	0,3	0,2	0:00.51	tracker-miner-f
2877	utfpr	20	0	3218252	75580	55820	S	0,3	0,5	0:01.17	gjs
2919	utfpr	20	0	961640	15940	12932	S	0,3	0,1	0:00.20	xdg-desktop-por
12961	root	20	0	0	0	0	I	0,3	0,0	0:00.44	kworker/u48:0-i915
14954	root	0	-20	0	0	0	I	0,3	0,0	0:00.26	kworker/u49:1-i915_flip
15022	utfpr	20	0	499988	35168	27732	S	0,3	0,2	0:00.09	gnome-control-c
1	root	20	0	24024	15456	9712	S	0,0	0,1	0:02.09	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/0:0H-events_highpri
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	kworker/0:1-i915-unordered
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq

Exercício C: Teste todas as opções do comando ps e também ps –aux, ps -u . Agora anote o que o comando ps faz, especialmente o que acontece quando você testa ps –aux e ps -u ?

```
utfpr@GP-LAB4-W1003:/home$ ps -e
```

PID	TTY	TIME	CMD
1	?	00:00:02	systemd
2	?	00:00:00	kthreadd
3	?	00:00:00	pool_workqueue_release
4	?	00:00:00	kworker/R-rcu_gp
5	?	00:00:00	kworker/R-sync_wq
6	?	00:00:00	kworker/R-kvfree_rcu_reclaim
7	?	00:00:00	kworker/R-slub_flushwq
8	?	00:00:00	kworker/R-netns
10	?	00:00:00	kworker/0:0H-events_highpri
13	?	00:00:00	kworker/R-mm_percpu_wq
14	?	00:00:00	ksoftirqd/0
15	?	00:00:03	rcu_preempt
16	?	00:00:00	rcu_exp_par_gp_kthread_worker/0
17	?	00:00:00	rcu_exp_gp_kthread_worker
18	?	00:00:00	migration/0
19	?	00:00:00	idle_inject/0
20	?	00:00:00	cpuhp/0
21	?	00:00:00	cpuhp/2
22	?	00:00:00	idle_inject/2
23	?	00:00:00	migration/2
24	?	00:00:00	ksoftirqd/2
26	?	00:00:00	kworker/2:0H-events_highpri
27	?	00:00:00	cpuhp/4
28	?	00:00:00	idle_inject/4
29	?	00:00:00	migration/4
30	?	00:00:00	ksoftirqd/4
32	?	00:00:00	kworker/4:0H-events_highpri
33	?	00:00:00	cpuhp/6
34	?	00:00:00	idle_inject/6
35	?	00:00:00	migration/6
36	?	00:00:00	ksoftirqd/6
38	?	00:00:00	kworker/6:0H-kblockd
39	?	00:00:00	cpuhp/8
40	?	00:00:00	idle_inject/8

```
utfpr@GP-LAB4-W1003:/home$ ps -l
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
0 S 1000 7707 7699 0 80 0 - 2169 do_wai pts/0 00:00:00 bash
4 S 1000 15001 14998 0 80 0 - 2169 do_wai pts/0 00:00:00 bash
4 R 1000 15578 15001 0 80 0 - 2825 - pts/0 00:00:00 ps
utfpr@GP-LAB4-W1003:/home$
```

```
utfpr@GP-LAB4-W1003:/home$ ps -a
PID TTY TIME CMD
2028 tty2 00:00:00 gnome-session-b
13942 pts/0 00:00:00 su
13944 pts/0 00:00:00 bash
14565 pts/0 00:00:00 su
14571 pts/0 00:00:00 bash
14998 pts/0 00:00:00 su
15001 pts/0 00:00:00 bash
15748 pts/0 00:00:00 ps
utfpr@GP-LAB4-W1003:/home$
```

```
utfpr@GP-LAB4-W1003:/home$ ps -u
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
utfpr 2017 0.0 0.0 233416 6680 tty2 Ssl+ 19:05 0:00 /usr/libexec/gdm-wayland-session env GNOME_SHELL_SESSION_MODE=ubuntu /usr/bin/gnome-session --session=ubuntu
utfpr 2028 0.0 0.1 295976 17240 tty2 Sl+ 19:05 0:00 /usr/libexec/gnome-session-binary --session=ubuntu
utfpr 7707 0.0 0.0 8676 5744 pts/0 Ss 19:21 0:00 bash
utfpr 15001 0.0 0.0 8676 5688 pts/0 S 19:40 0:00 bash
utfpr 15919 0.0 0.0 11268 4944 pts/0 R+ 19:44 0:00 ps -u
utfpr@GP-LAB4-W1003:/home$ ps -x
PID TTY STAT TIME COMMAND
1859 ? Ss 0:00 /usr/lib/systemd/systemd --user
1860 ? S 0:00 (sd-pam)
1870 ? S<sl 0:00 /usr/bin/pipewire
1871 ? Ssl 0:00 /usr/bin/pipewire -c filter-chain.conf
1874 ? S<sl 0:00 /usr/bin/wireplumber
1875 ? S<sl 0:00 /usr/bin/pipewire-pulse
1876 ? Sslsl 0:00 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/1000/keyring
1905 ? Ss 0:00 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
1968 ? Ssl 0:00 /usr/libexec/xdg-document-portal
1990 ? Ssl 0:00 /usr/libexec/xdg-permission-store
2017 tty2 Ssl+ 0:00 /usr/libexec/gdm-wayland-session env GNOME_SHELL_SESSION_MODE=ubuntu /usr/bin/gnome-session --session=ubuntu
2028 tty2 Sl+ 0:00 /usr/libexec/gnome-session-binary --session=ubuntu
2170 ? Ssl 0:00 /usr/libexec/xdg-desktop-portal-gnome
```

```
utfpr@GP-LAB4-W1003:/home$ ps -aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.0 24024 15456 ? Ss 19:05 0:02 /sbin/init splash
root 2 0.0 0.0 0 0 ? S 19:05 0:00 [kthreadd]
root 3 0.0 0.0 0 0 ? S 19:05 0:00 [pool_workqueue_release]
root 4 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-rcu_gp]
root 5 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-sync_wq]
root 6 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-kvfree_rcu_reclaim]
root 7 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-slub_flushwq]
root 8 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-netns]
root 10 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/0:0H-events_highpri]
root 13 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/R-mm_percpu_wq]
root 14 0.0 0.0 0 0 ? S 19:05 0:00 [ksoftirqd/0]
root 15 0.1 0.0 0 0 ? I 19:05 0:03 [rcu_preempt]
root 16 0.0 0.0 0 0 ? S 19:05 0:00 [rcu_exp_par_gp_kthread_worker/0]
root 17 0.0 0.0 0 0 ? S 19:05 0:00 [rcu_exp_gp_kthread_worker]
root 18 0.0 0.0 0 0 ? S 19:05 0:00 [migration/0]
root 19 0.0 0.0 0 0 ? S 19:05 0:00 [idle_inject/0]
root 20 0.0 0.0 0 0 ? S 19:05 0:00 [cpuhp/0]
root 21 0.0 0.0 0 0 ? S 19:05 0:00 [cpuhp/2]
root 22 0.0 0.0 0 0 ? S 19:05 0:00 [idle_inject/2]
root 23 0.0 0.0 0 0 ? S 19:05 0:00 [migration/2]
root 24 0.0 0.0 0 0 ? S 19:05 0:00 [ksoftirqd/2]
root 26 0.0 0.0 0 0 ? I< 19:05 0:00 [kworker/2:0H-events_highpri]
root 27 0.0 0.0 0 0 ? S 19:05 0:00 [cpuhp/4]
root 28 0.0 0.0 0 0 ? S 19:05 0:00 [idle_inject/4]
root 29 0.0 0.0 0 0 ? S 19:05 0:00 [migration/4]
```

```
utfpr@GP-LAB4-W1003:/home$ ps -u utfpr
```

PID	TTY	TIME	CMD
1859	?	00:00:00	systemd
1860	?	00:00:00	(sd-pam)
1870	?	00:00:00	pipewire
1871	?	00:00:00	pipewire
1874	?	00:00:00	wireplumber
1875	?	00:00:00	pipewire-pulse
1876	?	00:00:00	gnome-keyring-d
1905	?	00:00:00	dbus-daemon
1968	?	00:00:00	xdg-document-po
1998	?	00:00:00	xdg-permission-
2017	tty2	00:00:00	gdm-wayland-ses
2028	tty2	00:00:00	gnome-session-b
2170	?	00:00:00	mono
2196	?	00:00:00	gcr-ssh-agent
2197	?	00:00:00	gnome-session-c
2206	?	00:00:00	gvfsd
2222	?	00:00:00	gvfsd-fuse
2225	?	00:00:00	gnome-session-b
2266	?	00:00:00	at-spi-bus-laun
2268	?	00:02:06	gnome-shell
2280	?	00:00:00	dbus-daemon
2319	?	00:00:00	snapd-desktop-i
2394	?	00:00:00	at-spi2-registr
2395	?	00:00:00	snapd-desktop-i

Resposta: O comando ps tira uma "fotografia" dos processos em execução no exato momento em que você aperta Enter. Ao contrário do top, ele mostra a lista apenas uma vez e não fica atualizando a tela.

Esta é a combinação mais usada para diagnóstico, pois exibe todos os processos do sistema em detalhes:

- a: Mostra os processos de todos os usuários (não só os seus).
- u: Exibe informações detalhadas de consumo, como %CPU e %MEM, além de mostrar quem é o dono do processo.
- x: Inclui os processos de "fundo" do sistema (serviços que rodam silenciosamente sem janela de terminal).

O comando ps -u serve para filtrar a lista e mostrar apenas os processos de um usuário específico:

- Você deve informar o nome de quem deseja investigar logo à frente (ex: ps -u root listará apenas o que o administrador está rodando).
- Se digitado sozinho (dependendo do sistema), ele mostra os seus próprios processos de forma um pouco mais detalhada.

Exercício D: No terminal do Linux digite `man ps`. Agora veja que aparecem várias opções de uso do comando. Em especial teste os comandos abaixo e explique a saída deles.

\$ `ps aux`

\$ `ps axjf`

\$ `ps -U root -u root u`

\$ `ps -ao pid,user,cmd`

```
utfpr@GP-LAB4-W1003:/home$ man ps
utfpr@GP-LAB4-W1003:/home$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.0	24024	15456	?	Ss	19:05	0:02	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	19:05	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	19:05	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-rcu_gp]
root	5	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-sync_wq]
root	6	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-kvfree_rcu_reclaim]
root	7	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-slub_flushwq]
root	8	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-netns]
root	10	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/0:0H-events_highpri]
root	13	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-mm_percpu_wq]
root	14	0.0	0.0	0	0	?	S	19:05	0:00	[ksoftirqd/0]
root	15	0.1	0.0	0	0	?	I	19:05	0:04	[rcu_preempt]
root	16	0.0	0.0	0	0	?	S	19:05	0:00	[rcu_exp_par_gp_kthread_worker/0]
root	17	0.0	0.0	0	0	?	S	19:05	0:00	[rcu_exp_gp_kthread_worker]
root	18	0.0	0.0	0	0	?	S	19:05	0:00	[migration/0]
root	19	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/0]
root	20	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/0]
root	21	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/2]
root	22	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/2]
root	23	0.0	0.0	0	0	?	S	19:05	0:00	[migration/2]
root	24	0.0	0.0	0	0	?	S	19:05	0:00	[ksoftirqd/2]
root	26	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/2:0H-events_highpri]
root	27	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/4]

```
utfpr@GP-LAB4-W1003:/home$ ps axjf
```

PPID	PID	PGID	SID	TTY	TPGID	STAT	UID	TIME	COMMAND
0	2	0	0	?	-1	S	0	0:00	[kthreadd]
2	3	0	0	?	-1	S	0	0:00	_ [pool_workqueue_release]
2	4	0	0	?	-1	I<	0	0:00	_ [kworker/R-rcu_gp]
2	5	0	0	?	-1	I<	0	0:00	_ [kworker/R-sync_wq]
2	6	0	0	?	-1	I<	0	0:00	_ [kworker/R-kvfree_rcu_reclaim]
2	7	0	0	?	-1	I<	0	0:00	_ [kworker/R-slub_flushwq]
2	8	0	0	?	-1	I<	0	0:00	_ [kworker/R-netns]
2	10	0	0	?	-1	I<	0	0:00	_ [kworker/0:0H-events_highpri]
2	13	0	0	?	-1	I<	0	0:00	_ [kworker/R-mm_percpu_wq]
2	14	0	0	?	-1	S	0	0:00	_ [ksoftirqd/0]
2	15	0	0	?	-1	I	0	0:04	_ [rcu_preempt]
2	16	0	0	?	-1	S	0	0:00	_ [rcu_exp_par_gp_kthread_worker/0]
2	17	0	0	?	-1	S	0	0:00	_ [rcu_exp_gp_kthread_worker]
2	18	0	0	?	-1	S	0	0:00	_ [migration/0]
2	19	0	0	?	-1	S	0	0:00	_ [idle_inject/0]
2	20	0	0	?	-1	S	0	0:00	_ [cpuhp/0]
2	21	0	0	?	-1	S	0	0:00	_ [cpuhp/2]
2	22	0	0	?	-1	S	0	0:00	_ [idle_inject/2]
2	23	0	0	?	-1	S	0	0:00	_ [migration/2]
2	24	0	0	?	-1	S	0	0:00	_ [ksoftirqd/2]
2	26	0	0	?	-1	I<	0	0:00	_ [kworker/2:0H-events_highpri]
2	27	0	0	?	-1	S	0	0:00	_ [cpuhp/4]

```
utfpr@GP-LAB4-W1003:/home$ ps -U root -u root u
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.0	24024	15456	?	Ss	19:05	0:02	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	19:05	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	19:05	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-rcu_gp]
root	5	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-sync_wq]
root	6	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-kvfree_rcu_reclaim]
root	7	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-slub_flushwq]
root	8	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-netns]
root	10	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/0:0H-events_highpri]
root	13	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/R-mm_percpu_wq]
root	14	0.0	0.0	0	0	?	S	19:05	0:00	[ksoftirqd/0]
root	15	0.1	0.0	0	0	?	I	19:05	0:04	[rcu_preempt]
root	16	0.0	0.0	0	0	?	S	19:05	0:00	[rcu_exp_par_gp_kthread_worker/0]
root	17	0.0	0.0	0	0	?	S	19:05	0:00	[rcu_exp_gp_kthread_worker]
root	18	0.0	0.0	0	0	?	S	19:05	0:00	[migration/0]
root	19	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/0]
root	20	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/0]
root	21	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/2]
root	22	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/2]
root	23	0.0	0.0	0	0	?	S	19:05	0:00	[migration/2]
root	24	0.0	0.0	0	0	?	S	19:05	0:00	[ksoftirqd/2]
root	26	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/2:0H-events_highpri]
root	27	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/4]
root	28	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/4]
root	29	0.0	0.0	0	0	?	S	19:05	0:00	[migration/4]
root	30	0.0	0.0	0	0	?	S	19:05	0:00	[ksoftirqd/4]
root	32	0.0	0.0	0	0	?	I<	19:05	0:00	[kworker/4:0H-events_highpri]
root	33	0.0	0.0	0	0	?	S	19:05	0:00	[cpuhp/6]
root	34	0.0	0.0	0	0	?	S	19:05	0:00	[idle_inject/6]

```
utfpr@GP-LAB4-W1003:/home$ ps -ao pid,user,cmd
```

PID	USER	CMD
2028	utfpr	/usr/libexec/gnome-session-binary --session=ubuntu
13942	root	su pedro
13944	pedro	bash
14565	root	su madalena
14571	madalena	bash
14998	root	su utfpr
15001	utfpr	bash
17384	utfpr	ps -ao pid,user,cmd

```
utfpr@GP-LAB4-W1003:/home$
```

Resposta:

- ps aux: Mostra todos os processos do sistema (de todos os usuários) com informações detalhadas de consumo (CPU, memória, dono).
- ps axjf: Exibe todos os processos em formato de árvore hierárquica, facilitando ver quem é o "pai" e o "filho" de cada processo (qual programa abriu qual).
- ps -U root -u root u: Filtra e lista apenas os processos do usuário root (o administrador do sistema), mantendo o formato detalhado de exibição de colunas.
- ps -ao pid,user,cmd: Lista os processos de todos os usuários, mas exibe exatamente três colunas personalizadas: o ID do processo (pid), o dono (user) e o nome do programa/comando (cmd).

```
utfpr@GP-LAB4-W1003:/home$ pidof google chrome
14129 7565 4697 4570 4560 4395 4320 4305 3343 3151 3055 3046 3013 3011 3010 2992
```

Resposta: Ao utilizar o comando `pidof`, o terminal retornará apenas os números de identificação do processo.

Exercício E: Retorno dos comandos (`$ pidof mysqld` `$ pidof pico` `$ pidof firefox` `$ pidof gedit`)

```
utfpr@GP-LAB4-W1003:~$ pidof mysqld
utfpr@GP-LAB4-W1003:~$ pidof pico
utfpr@GP-LAB4-W1003:~$ pidof firefox
5324 5100
utfpr@GP-LAB4-W1003:~$ pidof gedit
utfpr@GP-LAB4-W1003:~$
```

Exercício F:

```
utfpr@GP-LAB4-W1003:~$ firefox &
[1] 6728
utfpr@GP-LAB4-W1003:~$ kill -STOP PID_FIREFOX
pidof firefox
5324 5100
[1]+  Concluido                firefox
utfpr@GP-LAB4-W1003:~$ kill -STOP PID_FIREFOX
bash: kill: PID_FIREFOX: argumentos devem ser IDs de trabalhos ou processo
utfpr@GP-LAB4-W1003:~$ kill -CONT PID_FIREFOX
bash: kill: PID_FIREFOX: argumentos devem ser IDs de trabalhos ou processo
utfpr@GP-LAB4-W1003:~$ kill -STOP PID_FIREFOX
bash: kill: PID_FIREFOX: argumentos devem ser IDs de trabalhos ou processo
utfpr@GP-LAB4-W1003:~$
```

Resposta: O navegador "congela" completamente, a interface deixa de responder aos seus cliques.

Exercício G:

```
utfpr@GP-LAB4-W1003:~$ touch resposta.txt
utfpr@GP-LAB4-W1003:~$ firefox &
[1] 10252
utfpr@GP-LAB4-W1003:~$ Ctrl-C
Gtk-Message: 18:07:17.010: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
mcopy resposta.txt resp mcopy resposta.txt resposta2.txt &
[2] 10898
utfpr@GP-LAB4-W1003:~$ Comando 'mcopy' não encontrado, mas poder ser instalado com:
sudo apt install mtools
ls *.txt
arq1.txt arq2.txt arq3.txt arquivoaps.txt resposta.txt seq1.txt seq2.txt telefones.txt Trabalho.txt
[2]+  Fim da execução com status 127      mcopy resposta.txt resposta2.txt
utfpr@GP-LAB4-W1003:~$ gedit
Comando 'gedit' não encontrado, mas poder ser instalado com:
sudo snap install gedit # version 48.1, or
sudo apt install gedit # version 46.1-3
Veja 'snap info gedit' para versões adicionais.
utfpr@GP-LAB4-W1003:~$ jobs
[1]+  Executando                firefox &
utfpr@GP-LAB4-W1003:~$
```

Exercício H:

```
utfpr@GP-LAB4-W1003:~$ jobs
[1]+  Executando                firefox &
utfpr@GP-LAB4-W1003:~$ gedit
Comando 'gedit' não encontrado, mas poder ser instalado com:
sudo snap install gedit # version 48.1, or
sudo apt install gedit # version 46.1.3
Veja 'snap info gedit' para versões adicionais.
utfpr@GP-LAB4-W1003:~$ sudo snap install gedit # version 48.1
[sudo] senha para utfpr:
gedit 48.1 from Canonical installed
utfpr@GP-LAB4-W1003:~$ gedit
libpbackend-1.0.so: cannot open shared object file: No such file or directory
Failed to load module: /home/utfpr/snap/gedit/common/.cache/gio-modules/libgiolibproxy.so
Gtk-Message: 20:13:38.941: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:12864): GLib-GIO-WARNING **: 20:13:38.941: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)
^Z
[2]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ jobs
[1]-  Executando                firefox &
[2]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ bg %1
bash: bg: o trabalho 1 já está em plano de fundo
utfpr@GP-LAB4-W1003:~$
```

Exercício I:

```
utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:38.371: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:31.409: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z[1]  Concluido                firefox

[3]+  Parado                    firefox
utfpr@GP-LAB4-W1003:~$ gedit
Gtk-Message: 20:17:23.428: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z
[4]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ jobs
[2]  Parado                    gedit
[3]-  Parado                    firefox
[4]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ bg %1
bash: bg: %1: trabalho não existe
utfpr@GP-LAB4-W1003:~$ bg %2
[2] gedit &
utfpr@GP-LAB4-W1003:~$
```

```
utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:38.371: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:31.409: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z[1]  Concluido                firefox

[3]+  Parado                    firefox
utfpr@GP-LAB4-W1003:~$ gedit
Gtk-Message: 20:17:23.428: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z
[4]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ jobs
[2]  Parado                    gedit
[3]-  Parado                    firefox
[4]+  Parado                    gedit
utfpr@GP-LAB4-W1003:~$ bg %1
bash: bg: %1: trabalho não existe
utfpr@GP-LAB4-W1003:~$ bg %2
[2] gedit &
utfpr@GP-LAB4-W1003:~$ pidof gedit
14226
[2]  Concluido                gedit
utfpr@GP-LAB4-W1003:~$ kill -STOP 14226
utfpr@GP-LAB4-W1003:~$ kill -CONT 14226
bash: kill: CONT14226: especificação de sinal inválida
utfpr@GP-LAB4-W1003:~$
```

```

utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:16.071: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Message: 20:16:51.496: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z[1] Concluido          firefox

[3]+  Parado              firefox

utfpr@GP-LAB4-W1003:~$ gedit
Gtk-Message: 20:17:23.420: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z
[4]+  Parado              gedit

utfpr@GP-LAB4-W1003:~$ jobs
[2]   Parado              gedit
[3]-  Parado              firefox
[4]+  Parado              gedit

utfpr@GP-LAB4-W1003:~$ bg %1
bash: bg: %1: trabalho não existe
utfpr@GP-LAB4-W1003:~$ bg %2
[2] gedit &
utfpr@GP-LAB4-W1003:~$ pidof gedit
14226
[2]   Concluido          gedit

utfpr@GP-LAB4-W1003:~$ kill -STOP 14226
utfpr@GP-LAB4-W1003:~$ kill -CONT14226
bash: kill: CONT14226: especificação de sinal inválida
utfpr@GP-LAB4-W1003:~$ kill -9 14226
utfpr@GP-LAB4-W1003:~$ idof firefox
Comando 'idof' não encontrado, você quis dizer:
  comando 'pidof' do deb sysvinit-utils (3.08-3ubuntu1)
Experimente: sudo apt install <deb name>
[4]+  Morto              gedit

utfpr@GP-LAB4-W1003:~$ pidof firefox
13789 13569
utfpr@GP-LAB4-W1003:~$ kill -9 13789 13569
utfpr@GP-LAB4-W1003:~$ █

```

```

utfrpr@GP-LAB4-W1003:~$ firefox &
[4] 14916
[3] Morto                firefox
utfrpr@GP-LAB4-W1003:~$ Gtk-Message: 20:21:25.008: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
gedit &
[5] 15549
utfrpr@GP-LAB4-W1003:~$ Gtk-Message: 20:21:41.008: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:15549): GLib-GIO-WARNING **: 20:21:41.008: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)
jobs
[4]: Executando          firefox &
[5]: Executando          gedit &
utfrpr@GP-LAB4-W1003:~$ fg %2
bash: fg: %2: trabalho nao existe
utfrpr@GP-LAB4-W1003:~$ killall -9 firefox
utfrpr@GP-LAB4-W1003:~$
utfrpr@GP-LAB4-W1003:~$ Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

```

Exercício J:

```

utfpr@GP-LAB4-W1003:~$ firefox
Gtk-Mensagem: 30/03/2024: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^Z
[6]+  Parado                  firefox
utfpr@GP-LAB4-W1003:~$ jobs
[5]-  Executando              gedit &
[6]+  Parado                  firefox
utfpr@GP-LAB4-W1003:~$ bg %1
bash: bg: %1: trabalho não existe
utfpr@GP-LAB4-W1003:~$ fg %1
bash: fg: %1: trabalho não existe
utfpr@GP-LAB4-W1003:~$ 

```

```
apoto
utfrpr@GP-LAB4-W1003:~$ ls /usr/local | more
bin
etc
games
include
lib
man
sbin
share
src
utfrpr@GP-LAB4-W1003:~$ ps -aux | more
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0  24156 15564 ?        Ss   19:19   0:02 /sbin/init splash
```

Exercício L:

```
top - 20:27:23 up 1:07, 2 users, load average: 0,42, 0,67, 0,59
Tarefas: 353 total, 1 em exec., 341 dormindo, 11 parado, 0 zumbi
%CPU(s): 1,3 us, 0,2 sy, 0,0 ni, 98,4 id, 0,1 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 6577,7 free, 3778,7 used, 6419,7 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used. 11901,5 avail mem
```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TEMPO+	COMANDO
8923	utfpr	20	0	1392,5g	612648	149644	S	10,6	3,8	5:20.59	chrome
2127	utfpr	20	0	5083084	378184	162044	S	4,6	2,4	3:50.77	gnome-shell
5022	utfpr	20	0	713504	75588	54276	S	1,0	0,5	0:20.00	gnome-terminal-
7539	utfpr	20	0	49,0g	212556	132300	S	1,0	1,3	0:52.74	chrome
2667	utfpr	20	0	419820	31268	19800	S	0,7	0,2	0:03.00	ibus-extension-
15	root	20	0	0	0	0	I	0,3	0,0	0:06.20	rcu_preempt
98	root	20	0	0	0	0	S	0,3	0,0	0:00.15	kcompactd0
2455	utfpr	20	0	386716	13184	7736	S	0,3	0,1	0:06.77	ibus-daemon
10083	root	20	0	0	0	0	I	0,3	0,0	0:01.28	kworker/u48:1-kvfree_rcu_reclaim
10781	root	0	-20	0	0	0	I	0,3	0,0	0:03.03	kworker/u49:2-i915_flip
1	root	20	0	24156	15564	9808	S	0,0	0,1	0:02.04	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/0:0H-events_highpri
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0,0	0,0	0:00.16	ksoftirqd/0
16	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.01	migration/0
19	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/2
22	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/2

```
utfpr@GP-LAB4-W1003:~$ nice -n 6 top
```

```
top - 20:28:06 up 1:08, 2 users, load average: 0,50, 0,67, 0,59
Tarefas: 360 total, 1 em exec., 348 dormindo, 11 parado, 0 zumbi
%CPU(s): 3,0 us, 0,9 sy, 0,1 ni, 95,8 id, 0,2 wa, 0,0 hi, 0,0 si, 0,0 st
MB mem : 15680,3 total, 6463,9 free, 3891,9 used, 6514,9 buff/cache
MB swap: 4096,0 total, 4096,0 free, 0,0 used. 11788,4 avail mem
```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TEMPO+	COMANDO
2127	utfpr	20	0	5075304	377604	162044	S	11,9	2,4	3:54.22	gnome-shell
17225	utfpr	20	0	3998540	168928	108896	S	8,9	1,1	0:00.27	gnome-character
17221	utfpr	20	0	1004984	56780	44384	S	5,6	0,4	0:00.17	nautilus
17227	utfpr	20	0	563676	50252	36748	S	5,0	0,3	0:00.15	gnome-clocks
1793	utfpr	20	0	11592	7380	4768	S	0,7	0,0	0:01.15	dbus-daemon
2455	utfpr	20	0	386716	13184	7736	S	0,7	0,1	0:06.89	ibus-daemon
2814	utfpr	39	19	946144	31048	19424	S	0,7	0,2	0:00.92	tracker-miner-f
7486	utfpr	20	0	48,8g	355452	258252	S	0,7	2,2	0:38.04	chrome
8923	utfpr	20	0	1392,5g	513448	153264	S	0,7	3,2	5:26.20	chrome
385	root	19	-1	67292	21620	19944	S	0,3	0,1	0:00.58	systemd-journal
1130	systemd+	20	0	17572	7784	6876	S	0,3	0,0	0:01.08	systemd-oomd
1311	syslog	20	0	222572	6584	4648	S	0,3	0,0	0:00.15	rsyslogd
2044	utfpr	20	0	754908	46700	26644	S	0,3	0,3	0:00.88	mono
2730	utfpr	20	0	234632	8336	7524	S	0,3	0,1	0:02.05	ibus-engine-sim
2870	utfpr	20	0	533616	9976	8496	S	0,3	0,1	0:00.05	gvfsd-trash
2885	utfpr	20	0	699608	15740	12712	S	0,3	0,1	0:00.34	xdg-desktop-por
5022	utfpr	20	0	713660	75624	54276	S	0,3	0,5	0:20.34	gnome-terminal-
7541	utfpr	20	0	48,4g	167628	102304	S	0,3	1,0	0:05.59	chrome
10781	root	0	-20	0	0	0	D	0,3	0,0	0:03.15	kworker/u49:2+i915_flip
14471	root	20	0	0	0	0	I	0,3	0,0	0:00.41	kworker/u48:3-i915
17175	root	20	0	0	0	0	I	0,3	0,0	0:00.04	kworker/u48:0-events_unbound
1	root	20	0	24156	15564	9808	S	0,0	0,1	0:02.05	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/0:0H-events_highpri
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0,0	0,0	0:00.16	ksoftirqd/0
15	root	20	0	0	0	0	I	0,0	0,0	0:06.27	rcu_preempt

```

utfpr@GP-LAB4-W1003:~$ gedit &
[8] 17375
utfpr@GP-LAB4-W1003:~$ Gtk-Message: 20:28:40.275: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
Firefox &
[9] 17456
[8] Concluido gedit
utfpr@GP-LAB4-W1003:~$ Gtk-Message: 20:28:41.700: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
ps -u
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
utfpr    1849  0.0  0.0 233416 6604 tty2    Ssl+ 19:20   0:00 /usr/libexec/gdm-wayland-session env GNOME_SHELL_SESSION_MODE=ubuntu /usr/bin/gnome-session
utfpr    1862  0.0  0.1 295976 17244 tty2    Sl+   19:20   0:00 /usr/libexec/gnome-session-binary --session=ubuntu
utfpr    5030  0.0  0.0  9748  6736 pts/0    Ss   19:20   0:00 bash
utfpr    15549  0.3  0.4 576068 72940 pts/0    Sl   20:21   0:01 /snap/gedit/697/usr/bin/gedit
utfpr    15915  0.7  3.0 1175764 497420 pts/0    TL   20:23   0:02 /snap/firefox/8339/usr/lib/firefox/firefox
utfpr    16137  0.0  0.2 449172 34732 pts/0    T    20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -ipcHandle 0 -signalPipe 1 -initial
utfpr    16140  0.0  0.2 462340 37516 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -parentBuildID 20260517181050 -pre
utfpr    16157  0.2  1.3 11105304 209340 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:38554
utfpr    16167  0.0  0.3 655816 63572 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -parentBuildID 20260517181050 -pre
utfpr    16256  0.0  0.7 2652524 119140 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:38554
utfpr    16389  0.0  0.2 461292 32672 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -parentBuildID 20260517181050 -sand
utfpr    16395  0.0  0.4 2599456 77692 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:46688
utfpr    16398  0.0  0.4 2599456 76580 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -parentBuildID 20260517181050 -sand
utfpr    16496  0.0  0.4 2599456 77200 pts/0    TL   20:23   0:00 /snap/firefox/8339/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:46688
utfpr    16784  0.0  0.0  6320  2976 pts/0    T    20:25   0:00 more
utfpr    17456  2.2  0.3 429172 57912 pts/0    Sl   20:28   0:00 /snap/firefox/8339/usr/lib/firefox/firefox
utfpr    17614  0.0  0.0 11396  5068 pts/0    R+   20:28   0:00 ps -u
utfpr@GP-LAB4-W1003:~$ renice -n 10 17375
renice: falha ao obter a prioridade para 17375 (process ID): Processo inexistente
utfpr@GP-LAB4-W1003:~$ renice -n 5 17375
renice: falha ao obter a prioridade para 17375 (process ID): Processo inexistente
utfpr@GP-LAB4-W1003:~$

```

```

utfpr@GP-LAB4-W1003:~$ pstree
systemd--ModemManager--3*[{ModemManager}]
--NetworkManager--3*[{NetworkManager}]
--accounts-daemon--3*[{accounts-daemon}]
--avahi-daemon--avahi-daemon
--bluetoothd
--colord--3*[{colord}]
--cron
--cups-browsed--3*[{cups-browsed}]
--cupsd
--dbus-daemon
--fwupd--5*[{fwupd}]
--gdm3--gdm-session-wor--gdm-wayland-ses--gnome-session-b--3*[{gnome-session-b}]
--3*[{gdm3}]--3*[{gdm-session-wor}]--3*[{gdm-wayland-ses}]
--gnome-remote-de--3*[{gnome-remote-de}]
--2*[{kerneloops}]
--mono--su--mono--11*[{mono}]
--12*[{mono}]
--mono--4*[{mono}]
--multipassd--dnsmasq
--19*[{multipassd}]
--polkitd--3*[{polkitd}]
--power-profiles- --3*[{power-profiles-}]
--rsyslogd--3*[{rsyslogd}]
--rtkit-daemon--2*[{rtkit-daemon}]
--snapd--18*[{snapd}]
--switcheroo-cont--3*[{switcheroo-cont}]
--systemd--(sd-pam)
--at-spi2-registr--3*[{at-spi2-registr}]
--chrome_crashpad--2*[{chrome_crashpad}]
--chrome_crashpad--chrome_crashpad
--2*[{crashhelper}--{crashhelper}]
--dbus-daemon
--dconf-service--3*[{dconf-service}]
--evolution-adre--6*[{evolution-adre}]
--evolution-calen--9*[{evolution-calen}]
--evolution-sourc--4*[{evolution-sourc}]
--gcr-ssh-agent--2*[{gcr-ssh-agent}]
--gnome-shell--11*[{gnome-shell}]

```

```
utfpr@GP-LAB4-W1003:~$ free -b
              total          usada       livre      compart.  buff/cache  disponível
Mem.:    16441946112  4212260864  6630490112  1003397120  6928543744  12229685248
Swap:      4294963200           0  4294963200
utfpr@GP-LAB4-W1003:~$ free -k
              total          usada       livre      compart.  buff/cache  disponível
Mem.:      16056588      4136808      6451780      1006972      6793280      11919780
Swap:       4194300           0      4194300
utfpr@GP-LAB4-W1003:~$ free -m
              total          usada       livre      compart.  buff/cache  disponível
Mem.:       15680         4007         6332           960         6611         11672
Swap:        4095           0         4095
utfpr@GP-LAB4-W1003:~$
```

Exercício M:

```
utfpr@GP-LAB4-W1003:~$ date
ter 23 jun 2026 20:33:10 -03
utfpr@GP-LAB4-W1003:~$ sudo date 1022180705 2021
[sudo] senha para utfpr:
date: operando extra "2021"
Tente "date --help" para mais informações.
utfpr@GP-LAB4-W1003:~$ cal
Comando 'cal' não encontrado, mas poder ser instalado com:
sudo apt install ncal
utfpr@GP-LAB4-W1003:~$ sudo apt install ncal
Lendo listas de pacotes... Pronto
Construindo árvore de dependências... Pronto
Lendo informação de estado... Pronto
Os seguintes pacotes foram instalados automaticamente e já não são necessários:
  libglapi-mesa python3-netifaces
Utilize 'sudo apt autoremove' para os remover.
Os NOVOS pacotes a seguir serão instalados:
  ncal
0 pacotes atualizados, 1 pacotes novos instalados, 0 a serem removidos e 265 não atualizados.
É preciso baixar 21,0 kB de arquivos.
Depois desta operação, 59,4 kB adicionais de espaço em disco serão usados.
Obter:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 ncal amd64 12.1.8 [21,0 kB]
Baixados 21,0 kB em 0s (130 kB/s)
A seleccionar pacote anteriormente não seleccionado ncal.
(Lendo banco de dados ... 242356 ficheiros e directórios atualmente instalados.)
A preparar para desempacotar .../archives/ncal_12.1.8_amd64.deb ...
A descompactar ncal (12.1.8) ...
Configurando ncal (12.1.8) ...
A processar 'triggers' para man-db (2.12.0-4build2) ...
utfpr@GP-LAB4-W1003:~$
```

```
utfpr@GP-LAB4-W1003:~$ cal
      Junho 2026
do se te qu qu se sá
   1  2  3  4  5  6
  7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30
utfpr@GP-LAB4-W1003:~$ date
ter 23 jun 2026 20:35:09 -03
utfpr@GP-LAB4-W1003:~$
```

Resumo:

Aqui está um resumo curto em formato de texto contínuo, ideal para introduzir o seu relatório de forma direta:

No Linux, todo programa em execução é tratado como um processo, recebendo um identificador único chamado PID. A interação com o sistema torna-se dinâmica pois os processos podem ser executados em primeiro plano (*foreground*), bloqueando o terminal, ou em segundo plano (*background*), adicionando o caractere & ao final do comando para liberar a linha de comando. O gerenciamento dessa transição é feito com ferramentas como o atalho CTRL+Z (que suspende um processo em andamento) e os comandos `jobs`, `bg` e `fg`, que permitem listar, enviar para o fundo ou trazer processos de volta para a tela principal, respectivamente.

Além do controle de tela, a administração do sistema exige a manipulação direta do estado dos programas. Isso é feito através de sinais enviados pelos comandos `kill` (via PID) e `killall` (via nome), que conseguem pausar, retomar ou forçar o encerramento imediato de aplicações. A eficiência do terminal também se apoia no uso de *Pipes* (`|`), um recurso poderoso que conecta a saída de um comando diretamente na entrada de outro, permitindo a execução de tarefas complexas e a filtragem de dados em uma única linha.

Por fim, o sistema operacional escalona o tempo de processador de cada tarefa através de um sistema de prioridades conhecido como *Nice* (NI). Utilizando os comandos `nice` (na criação) e `renice` (durante a execução), é possível ajustar a urgência de um processo. Valores negativos garantem alta prioridade e exigem privilégios de administrador, enquanto valores positivos indicam que o processo é "gentil", cedendo recursos da CPU para outras tarefas mais importantes do sistema.